

My title*

My subtitle if needed

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First sentence. Second sentence. Third sentence. Fourth sentence.

1 Introduction

Overview paragraph

Estimand paragraph

Results paragraph

Why it matters paragraph

Telegraphing paragraph: The remainder of this paper is structured as follows. Section 2....

2 Data

2.1 Overview

We use the statistical programming language R (R Core Team 2026).... Our data (Toronto Public Health 2026).... Following Alexander (2023), we consider...

Overview text

2.2 Measurement

Some paragraphs about how we go from a phenomena in the world to an entry in the dataset.

*Code and data are available at: https://github.com/UpCatch/Toronto_Tick_Surveillance_Data.

Table 1: The share of collected adult and nymph ticks that tested positive rose from under 6% before 2016 to between 29% and 43% from 2016 onward. No surveillance took place in 2020–2022.

Year	Parks surveyed	Parks with ticks	Adults and nymphs	Larvae	Tested positive	Positive (%)
2013	1	1	1	0	0	0.0
2014	4	1	3	0	0	0.0
2015	27	4	17	0	1	5.9
2016	36	6	95	0	30	31.6
2017	38	8	150	11	65	43.3
2018	46	12	96	11	36	37.5
2019	39	17	123	0	36	29.3
2023	42	8	121	0	43	35.5

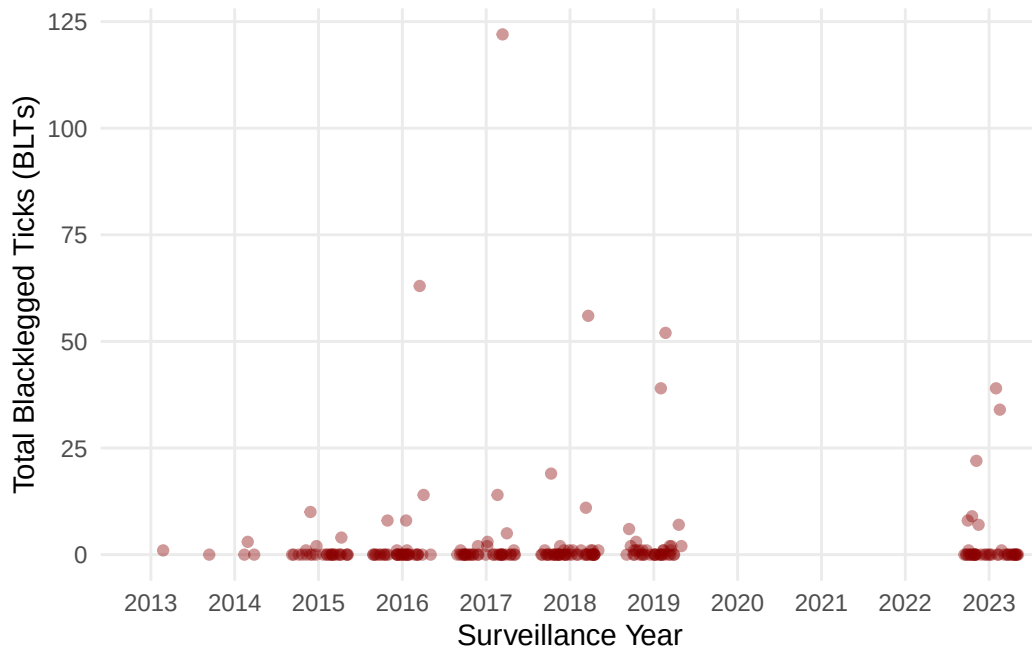


Figure 1: Distribution of total blacklegged ticks collected per park across surveillance years in Toronto.

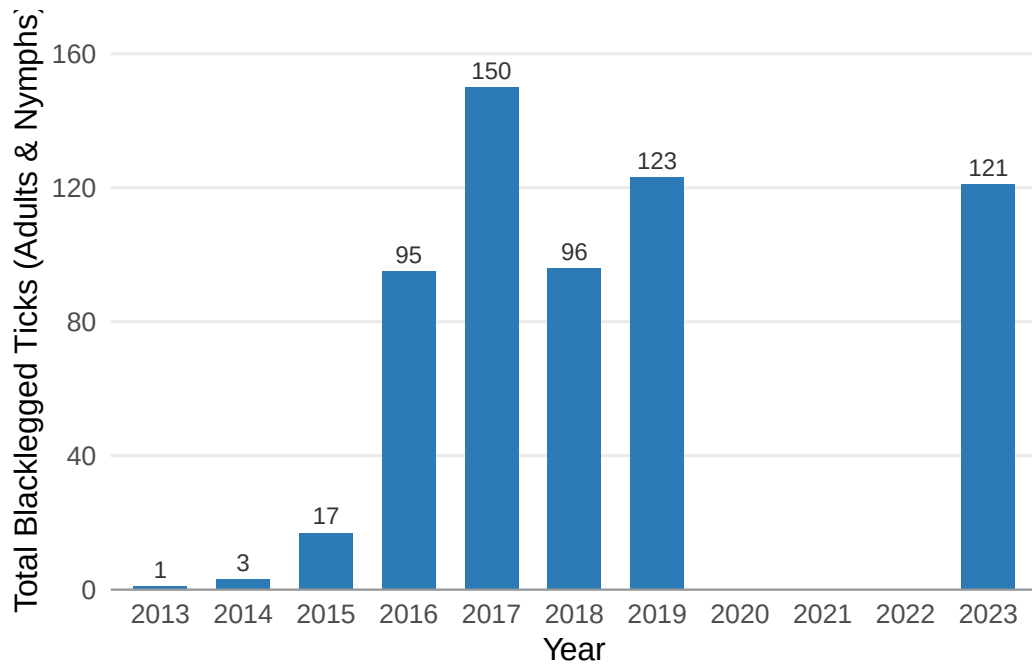


Figure 2: Total adult and nymph blacklegged ticks collected per year across all surveyed parks in Toronto.

References

- Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. <https://tellingstorieswithdata.com/>.
- R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>.
- Toronto Public Health. 2026. *Blacklegged Tick Surveillance*. <https://open.toronto.ca/dataset/blacklegged-tick-surveillance/>.

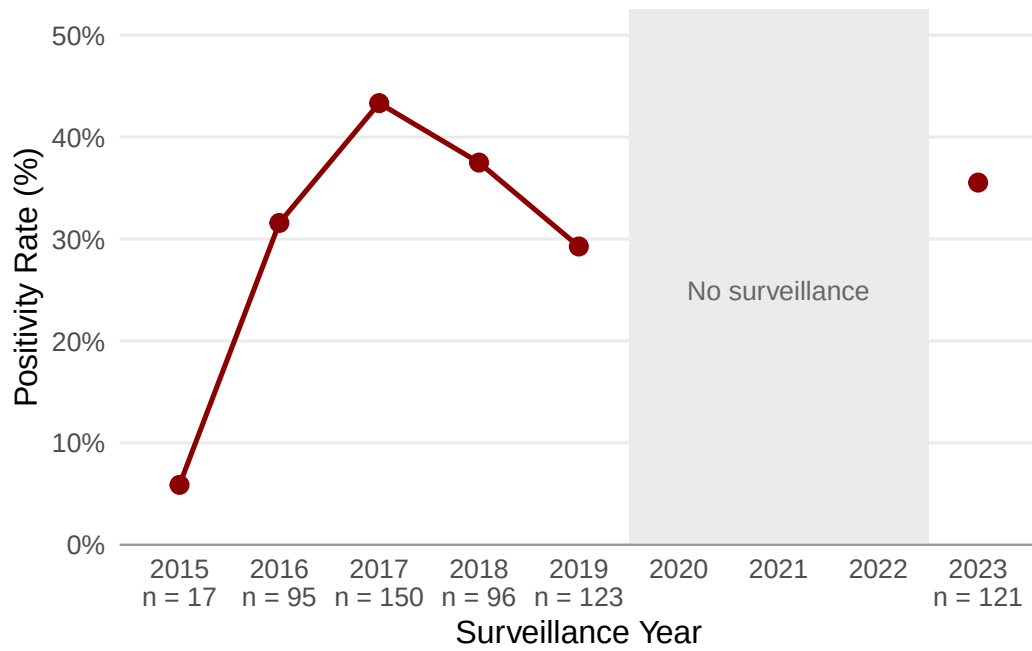


Figure 3: From 2016 onward, between 29% and 43% of the blacklegged ticks collected in Toronto parks each year tested positive. The number under each year (n) is the count of adult and nymph ticks collected that year. 2013 and 2014 are omitted because only one and three ticks were collected, too few for a meaningful rate. No surveillance took place in 2020–2022 (shaded).

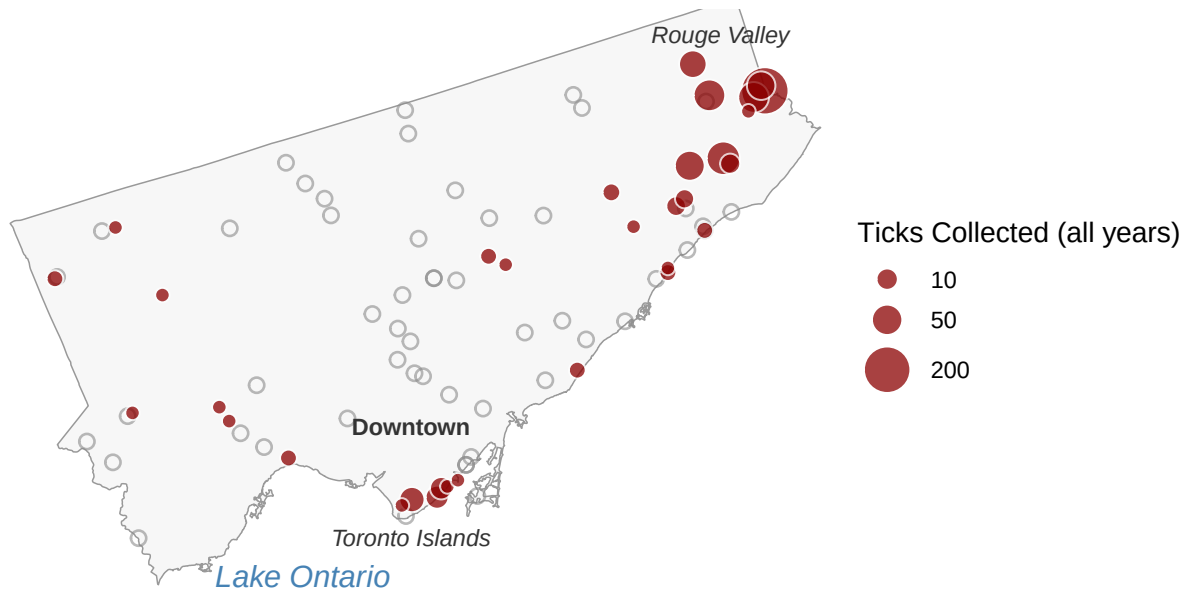


Figure 4: Blacklegged ticks cluster in the east end of Toronto, around the Rouge Valley, with a smaller cluster on the Toronto Islands. Each circle is one surveillance site, with counts summed over all years surveyed. Hollow grey circles mark sites where no ticks were ever found; red circles are sized by the total number of adult and nymph ticks collected. The grey outline is the City of Toronto boundary.

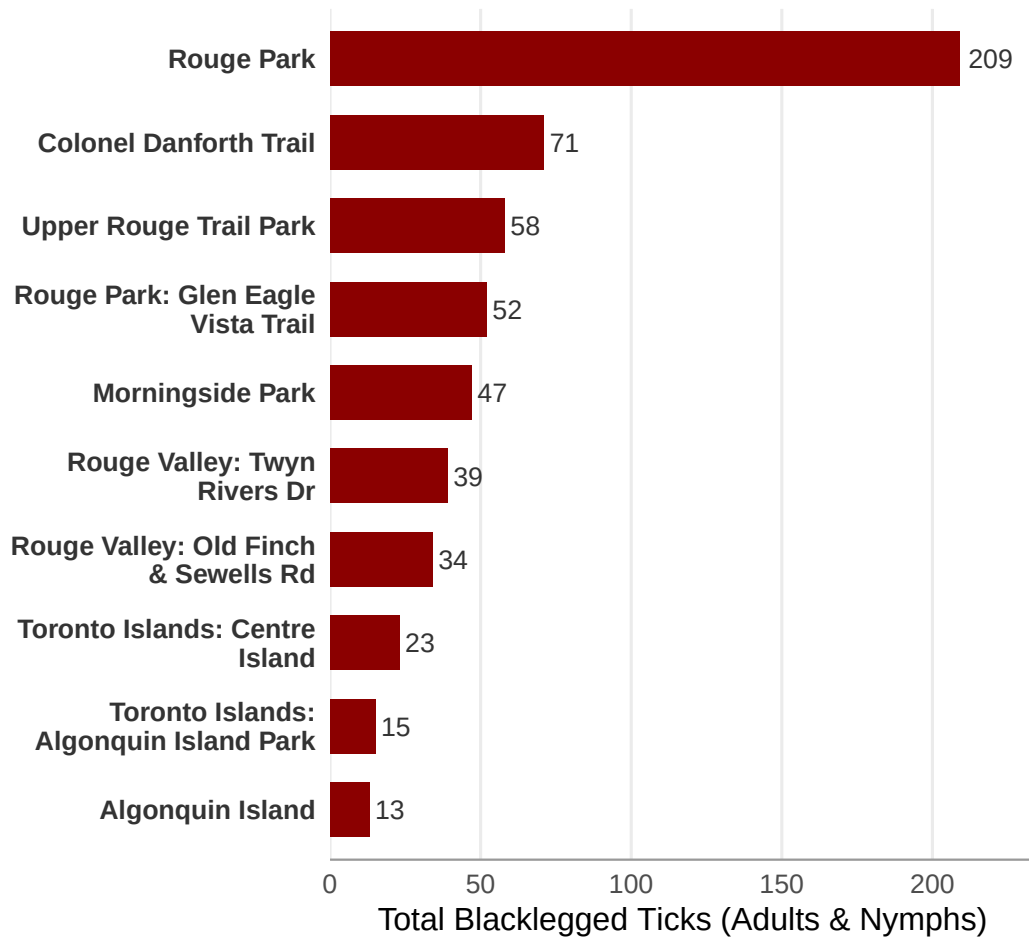


Figure 5: Top 10 Toronto parks with the highest cumulative blacklegged tick counts across all surveillance years.